HHI Report

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Overview

This analysis seeks to evaluate the relationship between an African country's Herfindahl-Hirschman Index (HHI) and their democracy score. The HHI measures the diversity of an African country's official development assistance gross (ODAG) total. Values near 1 indicate a country receiving nearly all of their aid from a single country whereas smaller values of HHI indicate a more balanced ODAG portfolio. We used democracy scores from Freedom House (TOTLIB1), the POLITY IV database (POLITY2), and the Liberal Democracy Index from the V-Dem Institute (V2x_Libdem). (Note: Here we only analyze POLITY2, whereas in previous reports we have analyzed both POLITY and POLITY2.)

This main analyses were split into a reduced and full model. The reduced model only considers democracy as a fixed effect, while including random intercepts for the African country and year. These random effects account for structural correlation between observations within the data structure.

The full model includes all of the parts of the reduced model while adding the African country's population, gross national income (GNI), total trade imports, total trade exports,and total ODAG (Note: When ODAG is not observed it is set to 0 in the analysis). Additionally, we included categorical variables for colonial past and region, and a cold war indicator. We applied a log transformation to GNI, population, total ODAG, imports, and exports due to their right skew. Quadratic and cubic transformations were added when necessary to capture nonlinear relationships. We applied this model framework to current year democracy scores as well as lagged democracy scores of 1, 2, and 3 years.

In the reduced models, every V2x_Libdem and POLITY2 coefficient is significant at $\alpha = 0.05$. The negative direction of their coefficients indicate that, as an African country is more democratic, their HHI drops on average meaning that they tend to have more external countries providing development aid and vice versa. Results for the significance of democracy variables in the reduced model are displayed in Tables 1-3.

In the full models, V2x_Libdem and TOTLIB are significant at the $\alpha = 0.05$ level for all lags. Conversely, POLITY2 is not significant at any lags. The negative coefficient for V2x_Libdem and the positive direction for TOTLIB1 yield the same conclusion as the the reduced models. That is, as an African country is more democratic, it tends to have lower HHI scores on average indicating a greater diversity of countries providing development aid and vice versa. Results for the significance of democracy variables in the reduced model are displayed in Tables 4-6. In addition to the democracy variables, population, GNI, and ODAG are all highly associated with HHI. The Cold War indicator is significant in the POLITY2 and v2x_Libdem models, but not in TOTLIB1 model. The Cold War indicator is positive indicating that during the Cold War HHI values were generally larger (i.e., less diverse) than after the Cold War ended.

Models

Reduced Model

The reduced model is described below, where the response Y_{it} (Y being HHI) ranges from $1/k$, k being the number of possible sources of ODAG, to 1 for African country i in year t. We applied a log transformation to HHI for better model fit. The predictor X_{1it} represents the democracy score of African country i in year t.

In lagged models, the year t changes units depending on the lag. For instance, a one-year lag would produce the model equation $\log(HHI_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$, where $X_{1i(t-1)}$ is the external country's democracy score in the prior year (year t-1).

$$Y_{it} \sim \mathrm{N}(\mu_{it}, \sigma^2)$$

$$\log(HHI_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$$

$$\gamma_{0i} \sim \mathrm{N}(0, \sigma^2_{country})$$

$$\gamma_{0t} \sim \mathrm{N}(0, \sigma^2_{year})$$

Full Model

Our full model equations are given below. They follow the same setup as the reduced model with the addition of the aforementioned transformed country characteristics. In lagged models, we only lagged the country's democracy score. For instance, a one-year lag would produce

the model equation $\log(HHI_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + ... + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$, where $X_{1i(t-1)}$ is the country's democracy score in the prior year (year t-1).

Additional predictors in this model include the African country's GNI, population, total ODAG, total imports and exports all on the log scale. Quadratic terms were included for the log-transformed GNI, population, and total ODAG. The log-transformed population variable added a cubic term to capture further nonlinear tendencies. In addition to these variables, colonial past, region, and a cold war indicator were included.

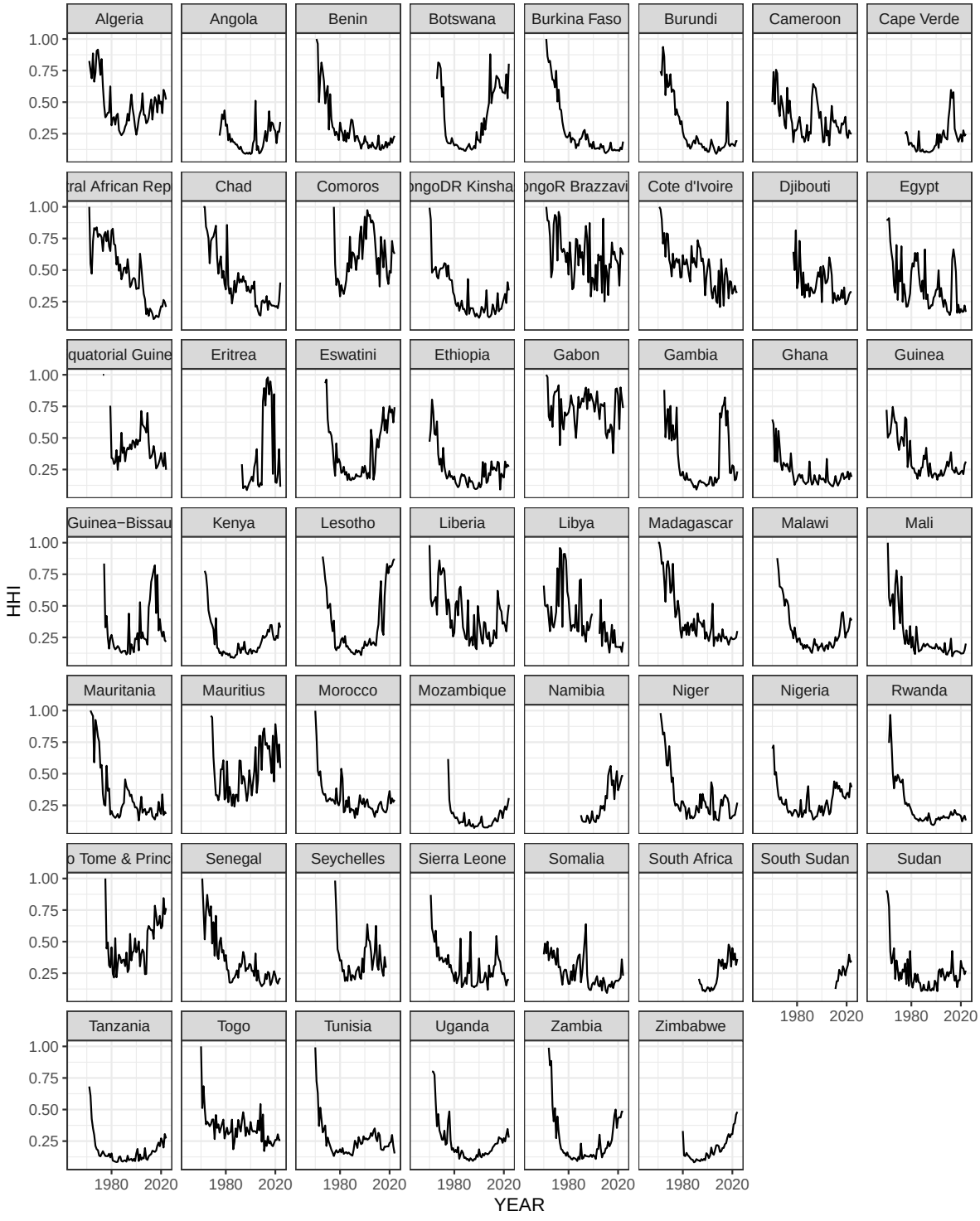
$$Y_{it} \sim \mathrm{N}(\mu_{it}, \sigma^2)$$

$$\log(HHI_{it}) = \beta_0 + \beta_1 X_{1ik} + ... + \gamma_{0i} + \gamma_{tk} + \epsilon_{it}$$

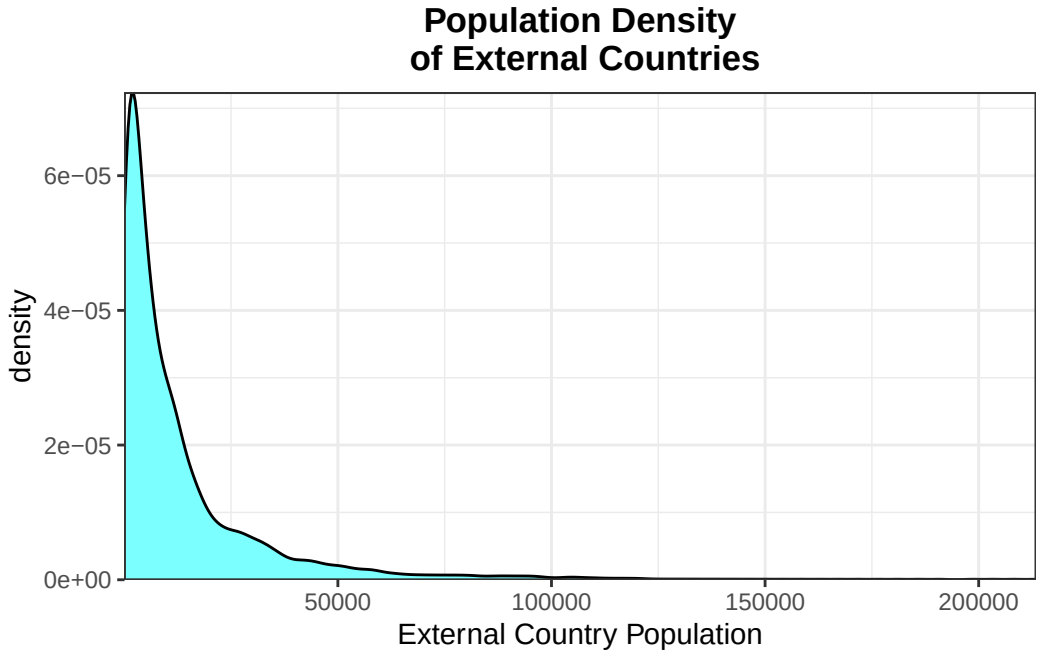
$$\gamma_{0i} \sim \mathrm{N}(0, \sigma^2_{country})$$

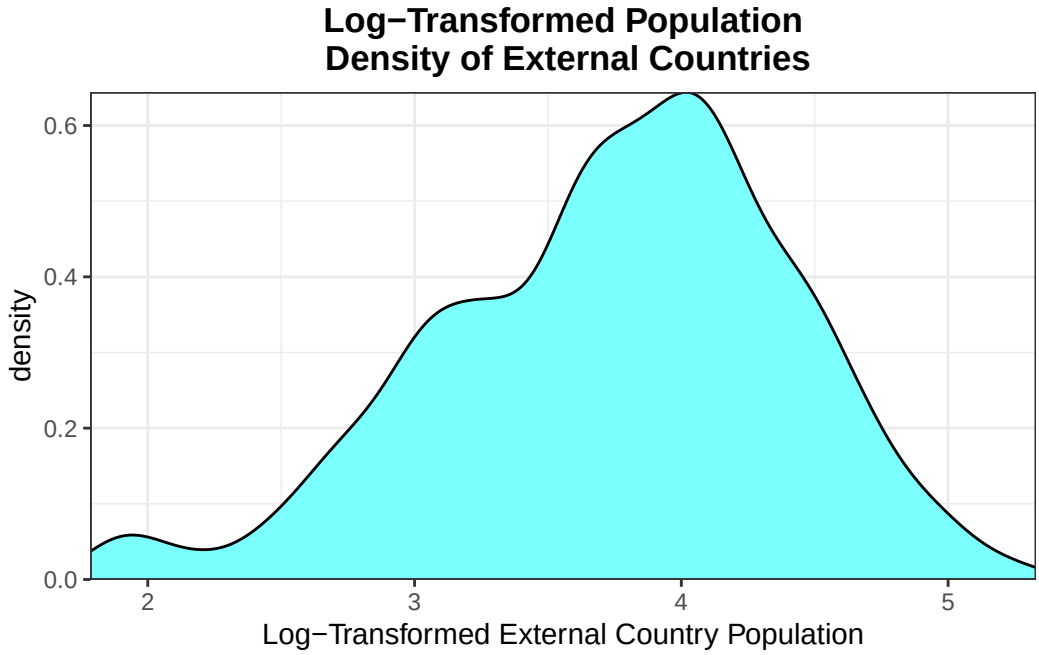
$$\gamma_{0t} \sim \mathrm{N}(0, \sigma^2_{year})$$

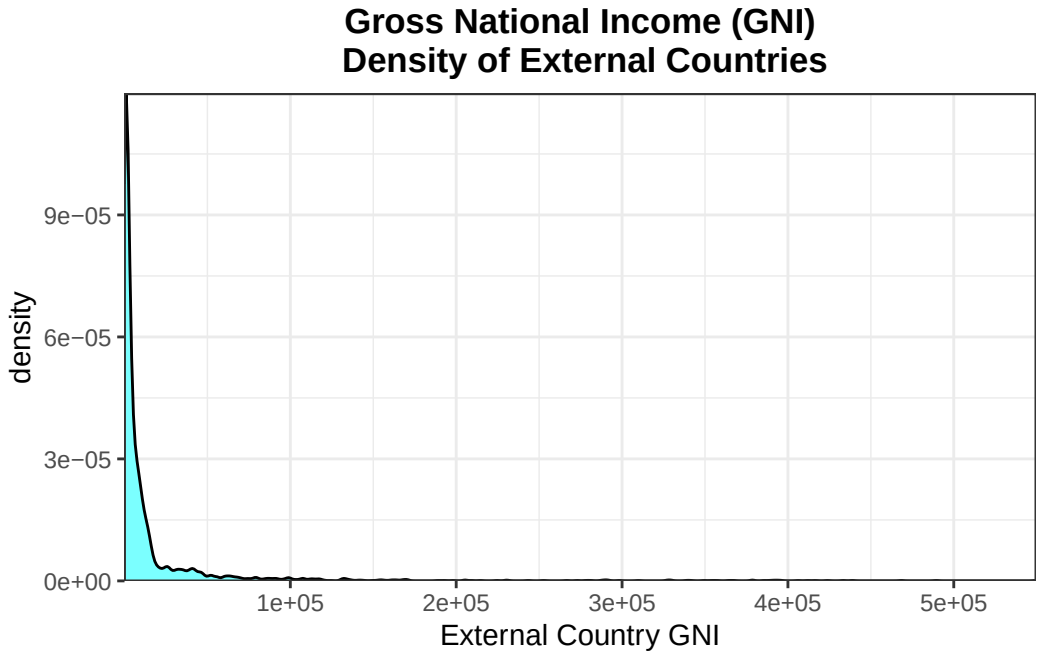
Data Visualizations and Results

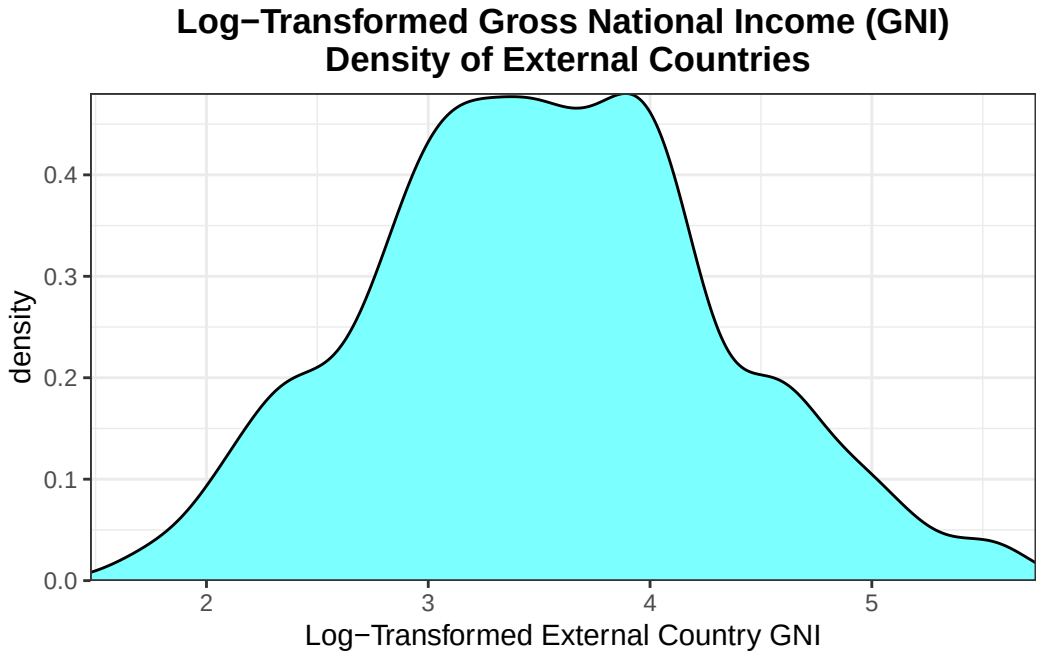


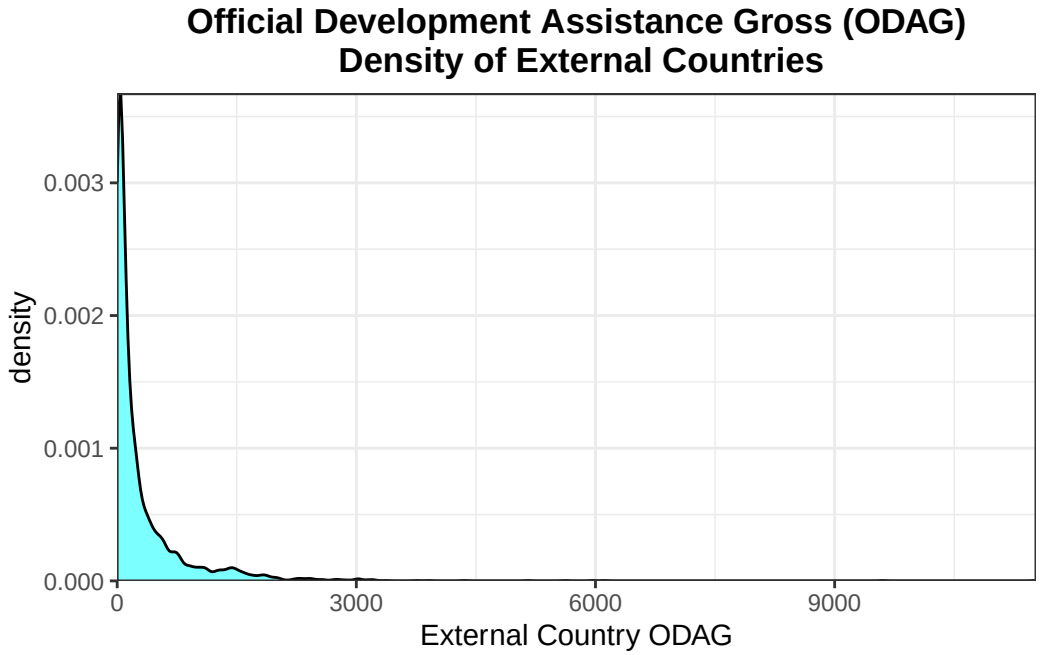
The distributions of the full model predictor variables and their log transformations are plotted below. The log-transformed variables all display a noticeably reduced right-skew, improving their performance and interpretation in the full models.

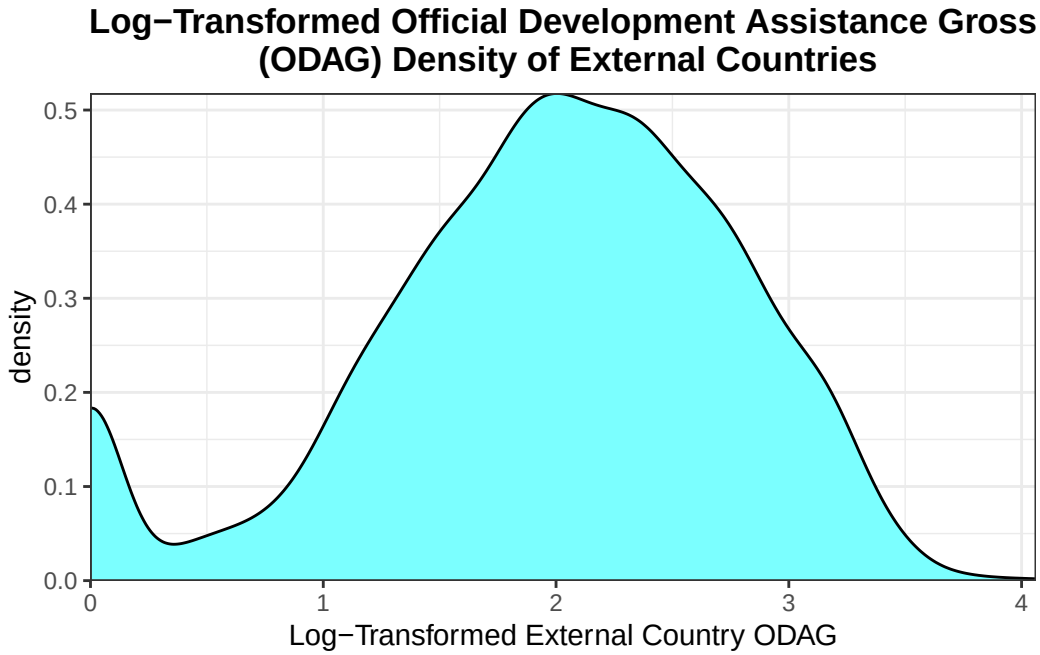





Gross National Income (GNI)
Density of External Countries
density
9e−05
6e−05
3e−05
0e+00
1e+05
2e+05
3e+05
4e+05
5e+05
External Country GNI


Log−Transformed Gross National Income (GNI)
Density of External Countries
density
0.4
0.3
0.2
0.1
0.0
2
3
4
5
Log−Transformed External Country GNI





The relationship between relevant democracy scores and log-transformed HHI, grouped and colored by year, and the relationship between log-transformed full model predictor variables and HHI are all visualized below.

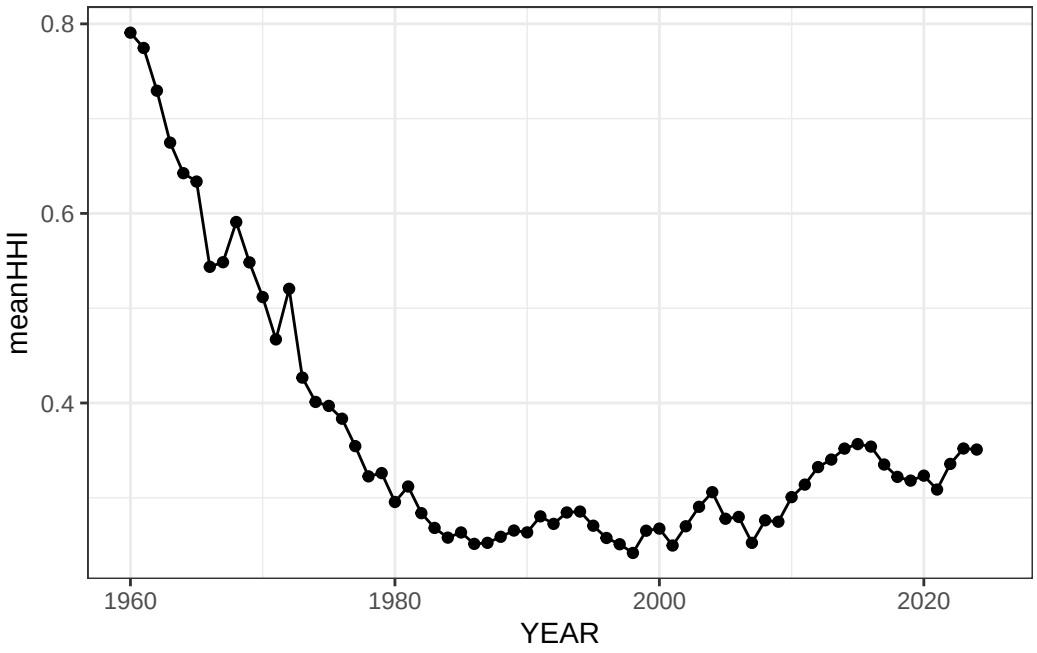
```
# A tibble: 54 x 2
   COUNTRY           TOTLIB1
   <chr>               <dbl>
 1 Somalia              13.8
 2 Equatorial Guinea    13.6
 3 South Sudan          13.3
 4 Libya                13.1
 5 Eritrea              13.0
 6 Sudan                12.9
 7 CongoDR Kinshasa     12.6
 8 Chad                 12.5
 9 Burundi              12.3
10 Angola               12.2
# i 44 more rows
```

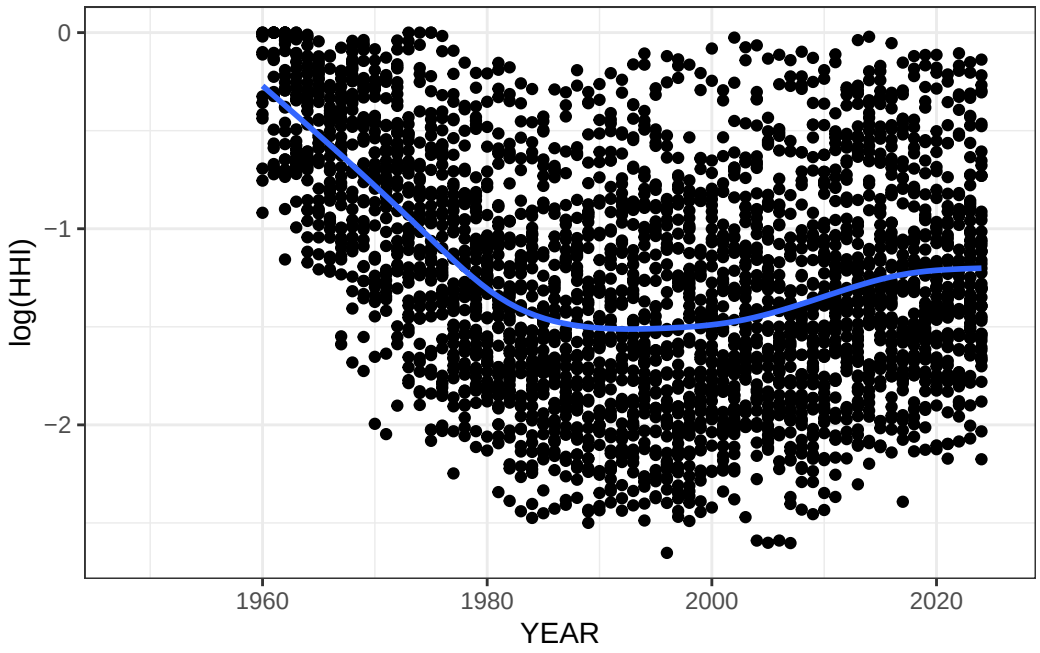
```
# A tibble: 54 x 2
   COUNTRY             V2x_Libdem
   <chr>                    <dbl>
 1 Mauritius                0.62
 2 Botswana                 0.536
 3 Namibia                  0.511
 4 Cape Verde               0.487
 5 Senegal                  0.453
 6 Sao Tome & Principe      0.350
 7 Ghana                    0.342
 8 South Africa             0.341
 9 Benin                    0.338
10 Seychelles               0.295
# i 44 more rows
```

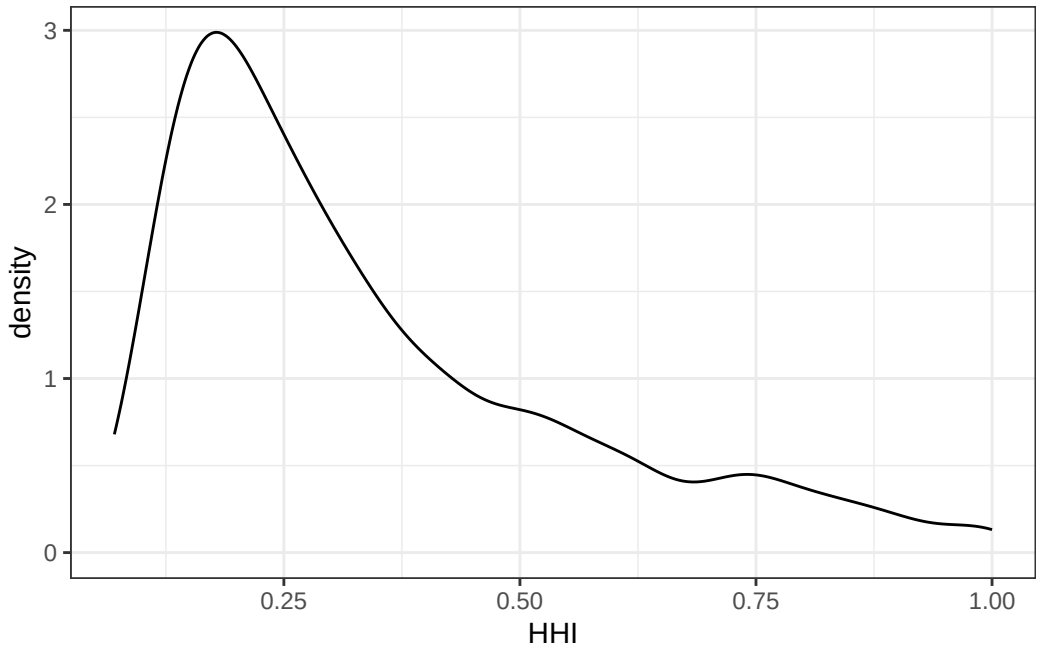
```
# A tibble: 52 x 2
   COUNTRY      POLITY2
   <chr>          <dbl>
 1 Mauritius      9.73
 2 Botswana       8.51
 3 Namibia        6.62
 4 South Africa   5.87
 5 Cape Verde     4.61
 6 Gambia         2.06
 7 Comoros        1.77
 8 Lesotho        1.08
 9 Madagascar     0.915
```

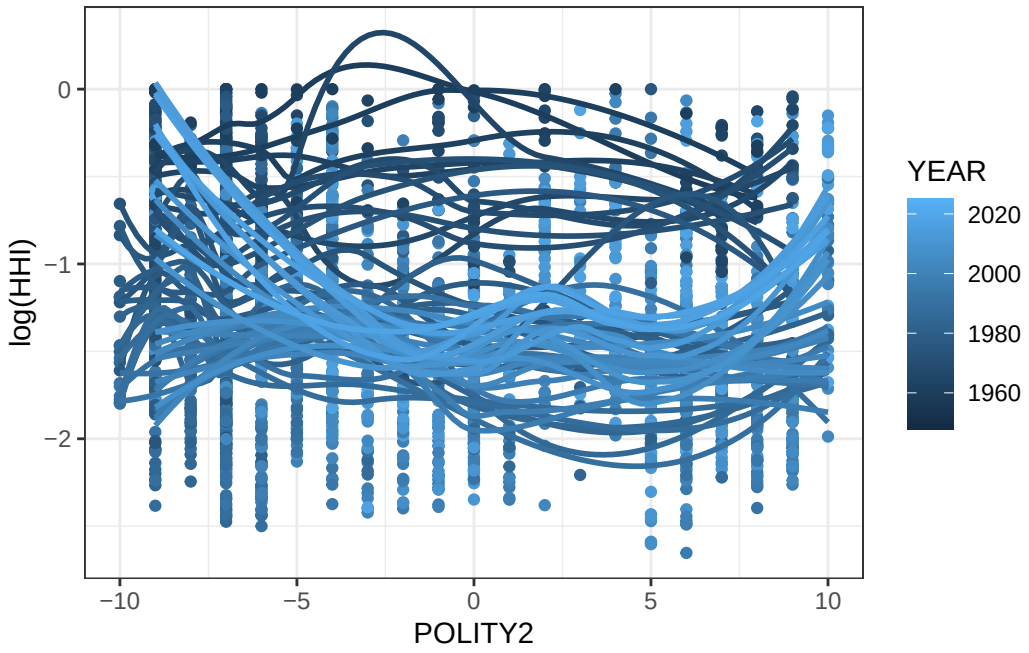
```
10 Benin          0.288
# i 42 more rows

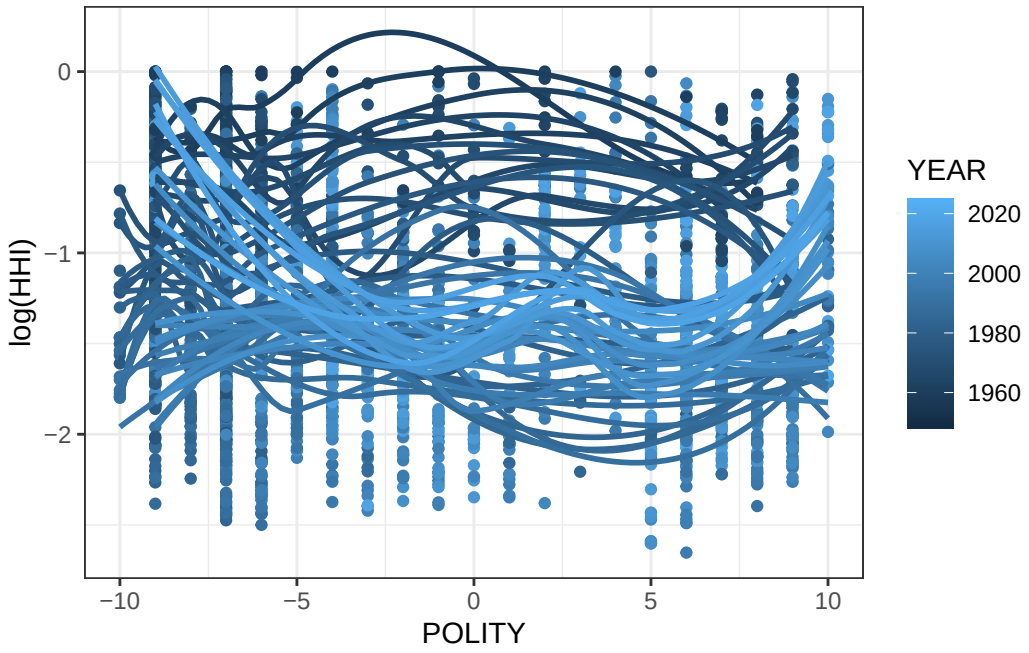

# A tibble: 66 x 2
    YEAR meanHHI
   <dbl>   <dbl>
 1  1960   0.791
 2  1961   0.775
 3  1962   0.729
 4  1963   0.675
 5  1964   0.642
 6  1965   0.634
 7  1966   0.544
 8  1967   0.549
 9  1968   0.591
10  1969   0.548
# i 56 more rows
```

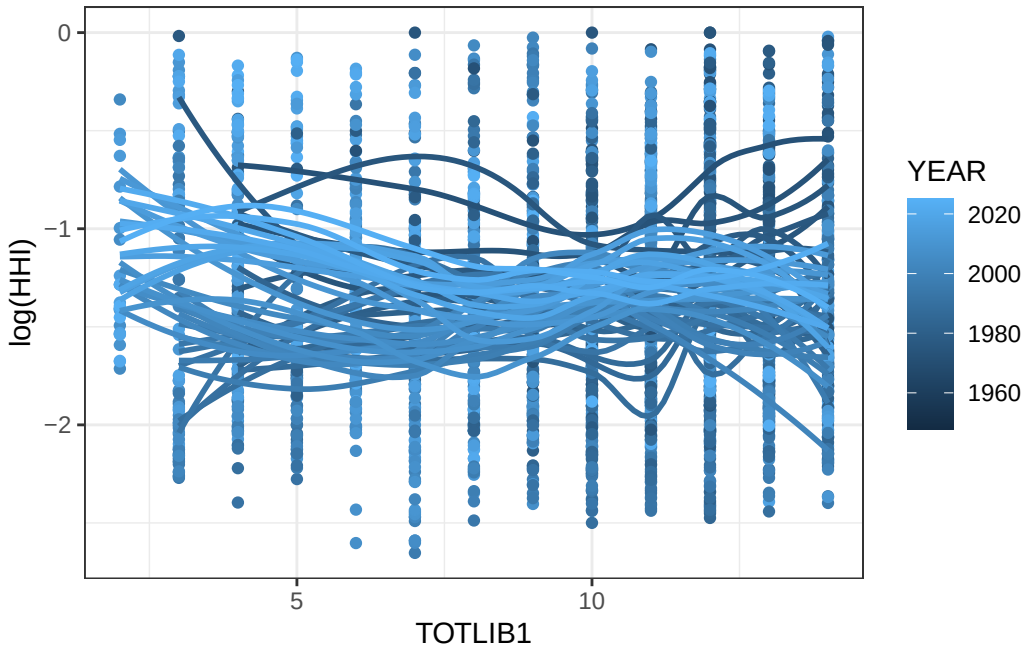


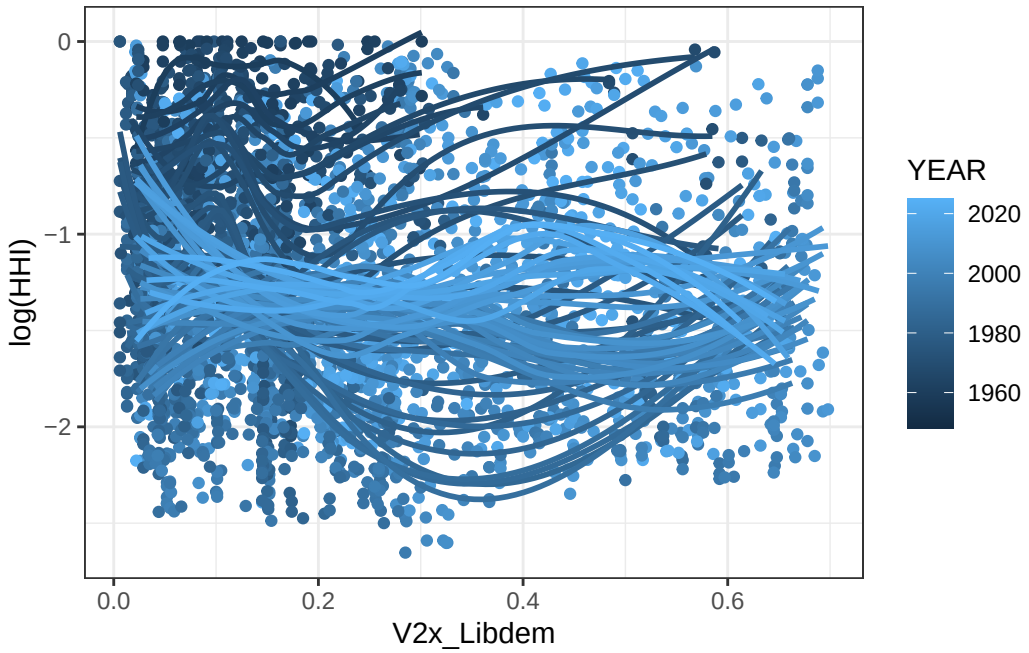

0
−1
−2
log(HHI)
1960
1980
2000
2020
YEAR

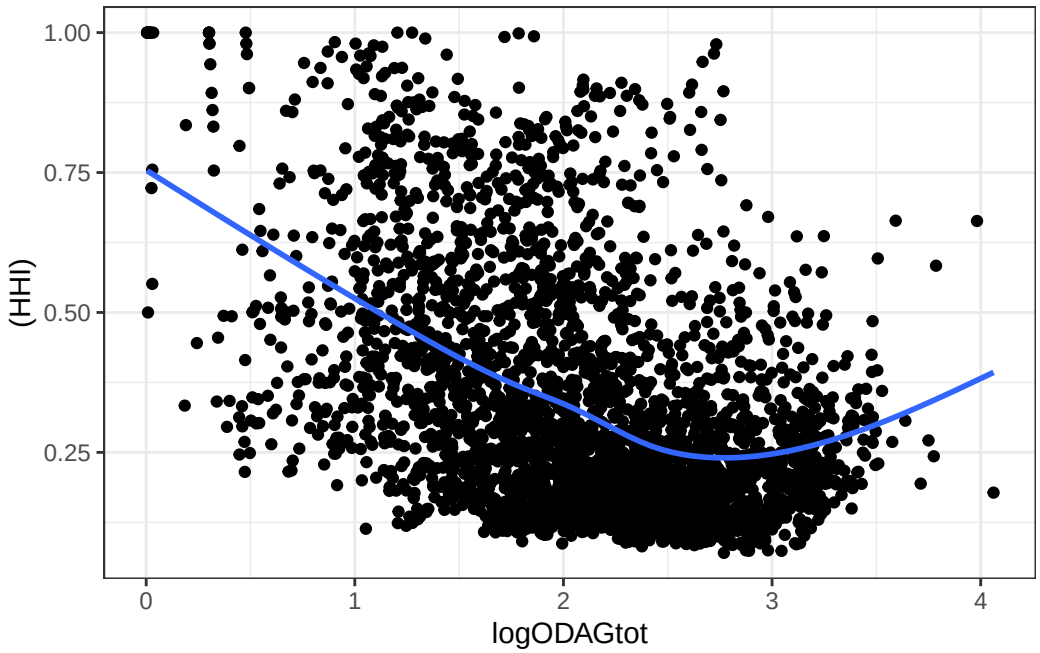

3
2
1
0
density
0.25
0.50
0.75
1.00
HHI


log(HHI)
0
−1
−2
−10
−5
0
5
10
POLITY2
YEAR
2020
2000
1980
1960


log(HHI)
0
−1
−2
−10
−5
0
5
10
POLITY
YEAR
2020
2000
1980
1960


0
−1
−2
log(HHI)
5
10
TOTLIB1
YEAR
2020
2000
1980
1960


0
−1
−2
log(HHI)
0.0
0.2
0.4
0.6
V2x_Libdem
YEAR
2020
2000
1980
1960


1.00
0.75
0.50
0.25
(HHI)
0
1
2
3
4
logODAGtot

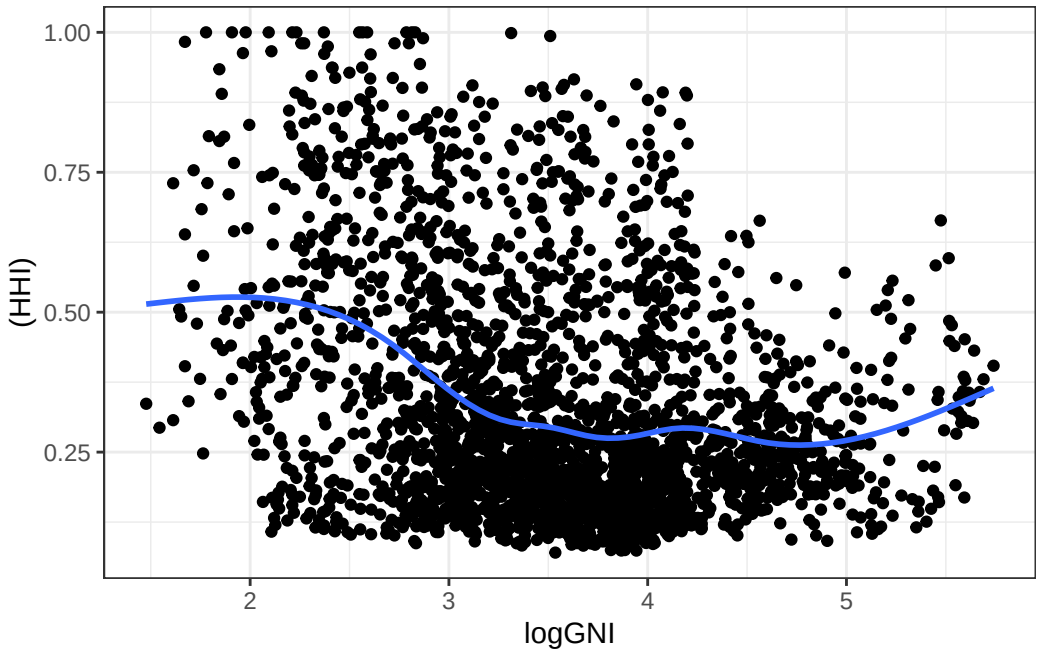
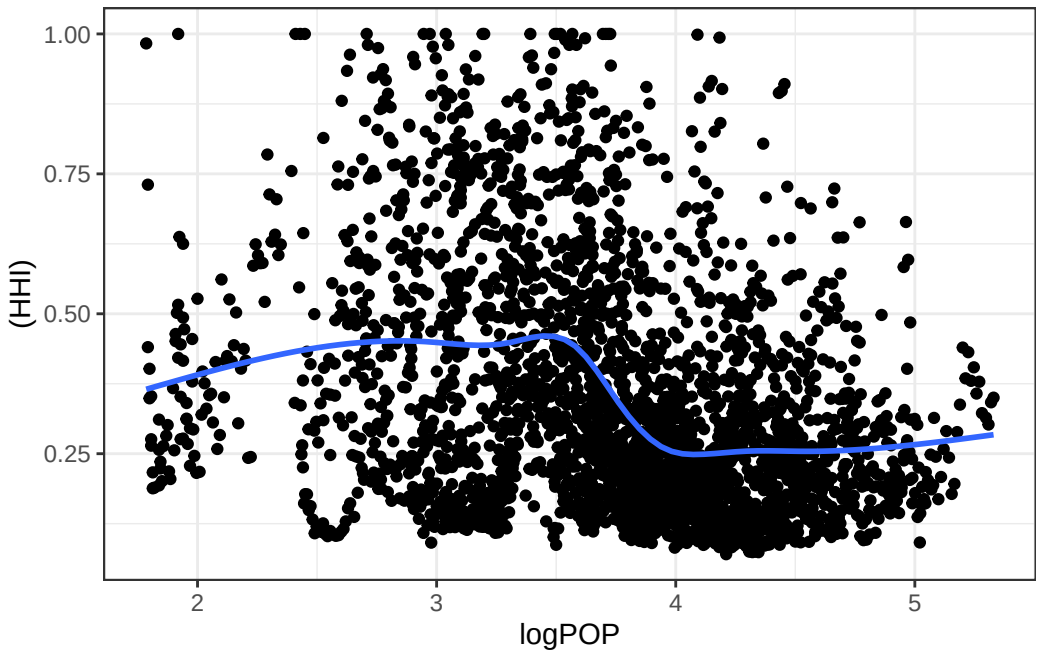

1.00
0.75
0.50
0.25
(HHI)
2
3
4
5
logGNI

Table 1: **Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.124***	0.067	−1.123***	0.067	−1.128***	0.066	−1.136***	0.066
V2x_Libdem	−0.326***	0.081						
V2x_Libdem_lag1			−0.358***	0.081				
V2x_Libdem_lag2					−0.363***	0.082		
V2x_Libdem_lag3							−0.351***	0.083
SD (Intercept YEAR)	0.355		0.346		0.336		0.327	
SD (Intercept CCODE)	0.349		0.350		0.353		0.356	
SD (Observations)	0.381		0.380		0.379		0.378	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001



Below are the summaries for the reduced models. The $\hat{\beta}$ coefficients are starred by significance as denoted in each table. Below each coefficient is their respective standard error.

Below are the summaries for the full models. As with the reduced model summaries, the $\hat{\beta}$ coefficients are starred by significance as denoted in each table, with their respective standard error below.

Table 2: **Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.222***	0.070	−1.231***	0.068	−1.239***	0.067	−1.243***	0.065
POLITY2	−0.006**	0.002						
POLITY2_lag1			−0.008***	0.002				
POLITY2_lag2					−0.008***	0.002		
POLITY2_lag3							−0.008***	0.002
SD (Intercept YEAR)	0.370		0.359		0.341		0.329	
SD (Intercept CCODE)	0.359		0.358		0.357		0.357	
SD (Observations)	0.375		0.375		0.375		0.376	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 3: **Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	−1.370***	0.067	−1.378***	0.066	−1.371***	0.066	−1.378***	0.066
TOTLIB1	0.003	0.004						
TOTLIB1_lag1			0.002	0.004				
TOTLIB1_lag2					0.001	0.004		
TOTLIB1_lag3							0.001	0.004
SD (Intercept YEAR)	0.168		0.143		0.133		0.126	
SD (Intercept CCODE)	0.367		0.369		0.371		0.372	
SD (Observations)	0.383		0.381		0.381		0.380	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 4: **Full Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−12.244***	1.623	−12.505***	1.631	−12.781***	1.646	−12.781***	1.664
logPOP	10.540***	1.322	10.733***	1.330	10.966***	1.343	10.946***	1.359
logPOP^2	−2.594***	0.354	−2.675***	0.356	−2.765***	0.360	−2.781***	0.365
logPOP^3	0.186***	0.032	0.196***	0.032	0.206***	0.032	0.209***	0.033
logGNI	−0.973***	0.141	−0.849***	0.143	−0.729***	0.146	−0.649***	0.148
logGNI^2	0.142***	0.019	0.126***	0.019	0.111***	0.019	0.100***	0.020
logODAGtot	−0.329***	0.081	−0.374***	0.085	−0.459***	0.088	−0.485***	0.096
logODAGtot^2	0.120***	0.019	0.130***	0.020	0.148***	0.021	0.153***	0.022
logTRDIMtot	0.055*	0.027	0.064*	0.027	0.065*	0.027	0.069*	0.028
logTRDEXtot	0.056*	0.025	0.055*	0.025	0.056*	0.026	0.053*	0.027
COL_British	0.126	0.163	0.119	0.161	0.106	0.160	0.095	0.158
COL_Portuguese	−0.113	0.227	−0.124	0.225	−0.135	0.224	−0.148	0.220
COL_Spanish	−0.371	0.449	−0.393	0.445	−0.421	0.441	−0.430	0.435
COL_Italian	−0.738*	0.330	−0.748*	0.327	−0.729*	0.324	−0.710*	0.320
COL_Belgian	0.327	0.456	0.310	0.452	0.297	0.448	0.278	0.442
COL_German	−0.275	0.207	−0.275	0.205	−0.274	0.203	−0.279	0.200
COL_US	−0.068	0.430	−0.065	0.426	−0.065	0.423	−0.058	0.416
COL_None	0.475	0.457	0.459	0.453	0.439	0.449	0.411	0.443
COL_Unique	−0.611	0.457	−0.609	0.453	−0.607	0.449	−0.611	0.443
REG_E_Africa	−0.161	0.242	−0.175	0.240	−0.176	0.238	−0.171	0.235
REG_S_Africa	−0.557*	0.262	−0.556*	0.260	−0.550*	0.258	−0.541*	0.254
REG_C_Africa	0.008	0.267	0.002	0.265	0.000	0.263	0.005	0.259
REG_W_Africa	−0.357	0.231	−0.357	0.229	−0.351	0.227	−0.349	0.224
COLD_WAR	0.212*	0.087	0.203*	0.086	0.192*	0.083	0.192*	0.082
SD (Intercept YEAR)	0.293		0.288		0.278		0.272	
SD (Intercept CCODE)	0.407		0.404		0.400		0.394	
SD (Observations)	0.347		0.345		0.343		0.343	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 5: **Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−20.076***	2.109	−19.057***	2.089	−18.905***	2.077	−18.324***	2.069
POLITY2	0.000	0.002						
POLITY2_lag1			−0.003	0.002				
POLITY2_lag2					−0.004+	0.002		
POLITY2_lag3							−0.004*	0.002
logPOP	16.335***	1.694	15.499***	1.676	15.450***	1.663	15.019***	1.654
logPOP^2	−4.011***	0.442	−3.814***	0.437	−3.832***	0.433	−3.758***	0.430
logPOP^3	0.304***	0.039	0.288***	0.038	0.292***	0.038	0.289***	0.038
logGNI	−1.103***	0.155	−0.987***	0.156	−0.877***	0.157	−0.782***	0.158
logGNI^2	0.161***	0.021	0.143***	0.021	0.126***	0.021	0.113***	0.021
logODAGtot	−0.318***	0.084	−0.373***	0.088	−0.454***	0.093	−0.480***	0.102
logODAGtot^2	0.116***	0.020	0.126***	0.021	0.142***	0.021	0.146***	0.023
logTRDIMtot	0.038	0.028	0.049+	0.028	0.050+	0.028	0.057*	0.029
logTRDEXtot	0.056*	0.027	0.056*	0.027	0.062*	0.027	0.062*	0.027
COL_British	−0.047	0.137	−0.041	0.136	−0.038	0.137	−0.041	0.135
COL_Portuguese	−0.279	0.204	−0.281	0.202	−0.283	0.204	−0.299	0.201
COL_Spanish	0.026	0.374	−0.026	0.372	−0.080	0.374	−0.125	0.370
COL_Italian	−0.648*	0.272	−0.671*	0.270	−0.667*	0.272	−0.651*	0.269
COL_Belgian	0.259	0.375	0.260	0.373	0.270	0.375	0.261	0.371
COL_German	−0.336*	0.169	−0.326+	0.168	−0.315+	0.169	−0.312+	0.167
COL_US	−0.133	0.351	−0.121	0.349	−0.117	0.351	−0.108	0.347
COL_None	0.318	0.377	0.323	0.375	0.317	0.377	0.294	0.373
COL_Unique	−0.689+	0.386	−0.646+	0.381	−0.640+	0.383	−0.596	0.376
REG_E_Africa	−0.119	0.200	−0.131	0.199	−0.146	0.200	−0.137	0.198
REG_S_Africa	−0.439*	0.220	−0.427+	0.218	−0.427+	0.219	−0.404+	0.217
REG_C_Africa	−0.002	0.224	−0.011	0.223	−0.035	0.224	−0.031	0.221
REG_W_Africa	−0.271	0.192	−0.275	0.191	−0.291	0.191	−0.287	0.189
COLD_WAR	0.315***	0.090	0.285**	0.087	0.250**	0.082	0.241**	0.081
SD (Intercept YEAR)	0.303		0.291		0.274		0.268	
SD (Intercept CCODE)	0.330		0.329		0.331		0.327	
SD (Observations)	0.345		0.345		0.344		0.345	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 6: **Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−11.148***	1.700	−11.384***	1.719	−11.904***	1.756	−12.535***	1.801
logPOP	9.235***	1.420	9.290***	1.436	9.536***	1.465	9.927***	1.502
logPOP^2	−2.389***	0.383	−2.445***	0.388	−2.548***	0.396	−2.663***	0.406
logPOP^3	0.181***	0.035	0.189***	0.035	0.201***	0.036	0.211***	0.037
logGNI	−0.522***	0.157	−0.393*	0.160	−0.281+	0.163	−0.200	0.167
logGNI^2	0.088***	0.021	0.072***	0.021	0.058**	0.022	0.049*	0.022
logODAGtot	−0.431***	0.118	−0.363**	0.124	−0.217+	0.130	−0.194	0.135
logODAGtot^2	0.143***	0.026	0.131***	0.027	0.101***	0.029	0.097**	0.029
logTRDIMtot	0.117***	0.029	0.120***	0.029	0.117***	0.029	0.104***	0.030
logTRDEXtot	0.028	0.027	0.034	0.028	0.041	0.028	0.054+	0.028
COL_British	0.049	0.148	0.049	0.146	0.052	0.146	0.058	0.147
COL_Portuguese	−0.173	0.206	−0.179	0.204	−0.175	0.204	−0.157	0.205
COL_Spanish	−0.385	0.406	−0.382	0.402	−0.369	0.400	−0.386	0.402
COL_Italian	−0.576+	0.298	−0.574+	0.296	−0.544+	0.295	−0.534+	0.296
COL_Belgian	0.147	0.413	0.126	0.409	0.097	0.408	0.081	0.410
COL_German	−0.301	0.186	−0.297	0.185	−0.298	0.184	−0.298	0.185
COL_US	−0.011	0.389	−0.006	0.385	0.000	0.384	0.004	0.385
COL_None	0.334	0.414	0.321	0.410	0.294	0.409	0.276	0.411
COL_Unique	−0.617	0.415	−0.606	0.411	−0.588	0.409	−0.591	0.411
REG_E_Africa	−0.094	0.219	−0.097	0.216	−0.088	0.216	−0.068	0.217
REG_S_Africa	−0.421+	0.237	−0.413+	0.235	−0.411+	0.234	−0.414+	0.235
REG_C_Africa	0.142	0.242	0.148	0.239	0.156	0.238	0.163	0.239
REG_W_Africa	−0.248	0.209	−0.237	0.207	−0.230	0.206	−0.222	0.207
COLD_WAR	0.069	0.057	0.042	0.050	0.029	0.047	0.014	0.045
SD (Intercept YEAR)	0.159		0.130		0.119		0.107	
SD (Intercept CCODE)	0.367		0.363		0.361		0.363	
SD (Observations)	0.345		0.343		0.342		0.343	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001